Dr. Elizabeth J. French's Combination Battery, No. 4, is similar to No. 3, but has her latest patent, the silver helix in addition.

CARE OF THE BATTERY.

The solution for the batteries consists of one and three-quarters (1¾) ounces bichromate of potassa, three-quarters (¾) of an ounce of saltpetre dissolved in twenty-seven (27) ounces of hot water, in an earthenware vessel. When cold, add three (3) fluid-ounces commercial sulphuric acid, and one-quarter (¼) ounce bisulphate of mercury (avoirdupois).

In filling and emptying the cell do not remove the top; as doing so gives needless trouble and renders the carbons liable to be broken. Use the central aperture. Do not fill the cell above the lower part of the curve of the neck, and never use hot or warm solution.

Caution.—The battery-fluid will eat and destroy any fabric with which it comes in contact, unless immediately sponged out with pure water, and the material moistened in a solution of aqua ammonia, which will restore the color. It is wise to keep a small bottle in the case. The person using the battery should avoid breathing the emanations which are emitted from the open-cup.

TO START THE BATTERY.

Place the cell in its receptacle, immerse the zinc plate with its connecting clamp through the aperture, and connect the clamp and the binding-post of the carbons on the left with the two binding-posts